

# **Environmental Impact Assessment of Pollutant Emissions from Eraring Power Station, NSW**

**Environmental Pollution**

**ENVS8205**

**Group 04**

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## **Group Report Contributions**

Introduction, Objectives and Conclusion - Md Fahim Rahman

Methods- Md Minhazur Rehman

Results and Discussion – Muhammad Azeem Mazhar

Study Area and Recommendations - Minhazul Abedin

## **Abstract**

The use of coal to produce power continues to be one of the major contributors of greenhouse gases and air pollutants in the world, and its negative effects on the environment and human health are deleterious (Farrow et al. 2020; WHO 2021). The present paper assesses the emission profile, dispersion and impact profile of the Eraring power station which is the largest coal-fired power plant in Australia and is situated in the western bank of Lake Macquarie in the state of NSW. Based on the verified data sets of the National Pollutant Inventory (NPI) and Clean Energy Regulator (CER) employed in 2023/24, coupled with ReCiPe2016 Midpoint (H) life-cycle modelling applied to OpenLCA, the report estimates the volume of major pollutants, such as CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, PM<sub>2.5</sub>, and Hg, and determines the level of their impact on the environment and health. Findings reveal that Eraring released 13.55 0.868 t CO<sub>2</sub>-eq MWh<sup>-1</sup> which is more than the national mean of the generation, by a margin of about 40 percent. The dispersion of pollutants by modelling indicates that the facility never fails to exceed NEPM air-quality standards on occasions when there are the events of thermal inversion. These results highlight the need to strengthen emission control measures, establish monitoring and policy alignment with the best practices as the international practices.

## **Introduction**

### **Importance of the Problem**

The coke-fired power plants are a huge source of anthropogenic contamination and greenhouse gases in the atmosphere. Globally, over 75 per cent of CO<sub>2</sub> and 90 per cent of energy-related particulate pollutants are due to fossil-fuel burning (IPCC 2022). In Australia, coal remains the primary provider of electricity, providing more than half the total power needs, with almost a third of the total greenhouse gas emissions in the country, although there is a slow shift towards renewable energy (Clean Energy Regulator 2024). Such emissions discharge the collection of pollutants, including CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, PM<sub>2.5</sub>, VOCs, and trace metals, like Hg and As, which worsen the quality of the air, trigger acid deposition, and generate chronic respiratory diseases (Romanello et al. 2022; WHO 2021).

### **Background and Literature Review**

The previous studies have defined electricity generation via coal as one of the biggest contributors to air pollution in Australia, specifically, in the NSW Hunter-Lake Macquarie industrial belt (Farrow et al. 2020; Eguchi et al. 2021). The Lethal Power report by the Greenpeace organization in Australia Pacific revealed that emissions of the top 5 NSW stations, which include Eraring, Bayswater, Liddell, Vales Point and Mt Piper, are linked to more than 477 and 7,500 deaths and cases of asthma annually respectively (Farrow et al. 2020). Life-cycle analysis also shows that coal-based systems have the highest potential of global warming of up to 900 kg CO<sub>2</sub>-e per MWh, compared to less than 100 kg CO<sub>2</sub>-e in gas and renewable systems (Tang and Mudd 2015; Asif et al. 2022). These sources of PM<sub>2.5</sub> are associated with cardiovascular and respiratory illnesses and

higher mortality (Pope et al. 2015). The largest black-coal power plant in the country belongs to Origin Energy Limited and is called the Eraring Power Station with the generation capacity of 2,880 MW. It is located on the west side of Lake Macquarie and has been in operation since 1982 and has been an important part of the NSW electricity grid. Nevertheless, it is an old, sub-critical boiler, with no modern desulfurization and NO<sub>x</sub>-reduction systems, which makes it one of the biggest emitters in the region (EPA NSW 2023).

## **Aims and Objectives**

This research aims at assessing the emission profiles and environmental effects of the Eraring Power Station against the Australian average electricity generation profile.

The targeted objectives are to:

- Measure the greenhouse gas and priority air pollutant emissions by Quantify Eraring in the annual reports in NPI and CER data.
- OpenLCA Model pollutant dispersion and deposition patterns with ReCiPe 2016 Midpoint (H).
- Evaluate possible human health and ecological effects.
- Advise on emission control and policy measures to bring the activities of Eraring in line with the sustainable energy targets.

## **Study Area**

Eraring Power Station is a black coal fired power station. It located in NSW on the shore of Macquarie lake, approximately 120km north of Sydney, 40km south of Newcastle ( NSW Department of Planning and Environment, 2022)

The station is comprises four 720 MW coal fired generator units and one 42 MW diesel generator. All those give the power station generating capacity of 2922 MW and makes Australia's largest power station. Also, this power station accounts for approximately 25% of NSW's power requirements.

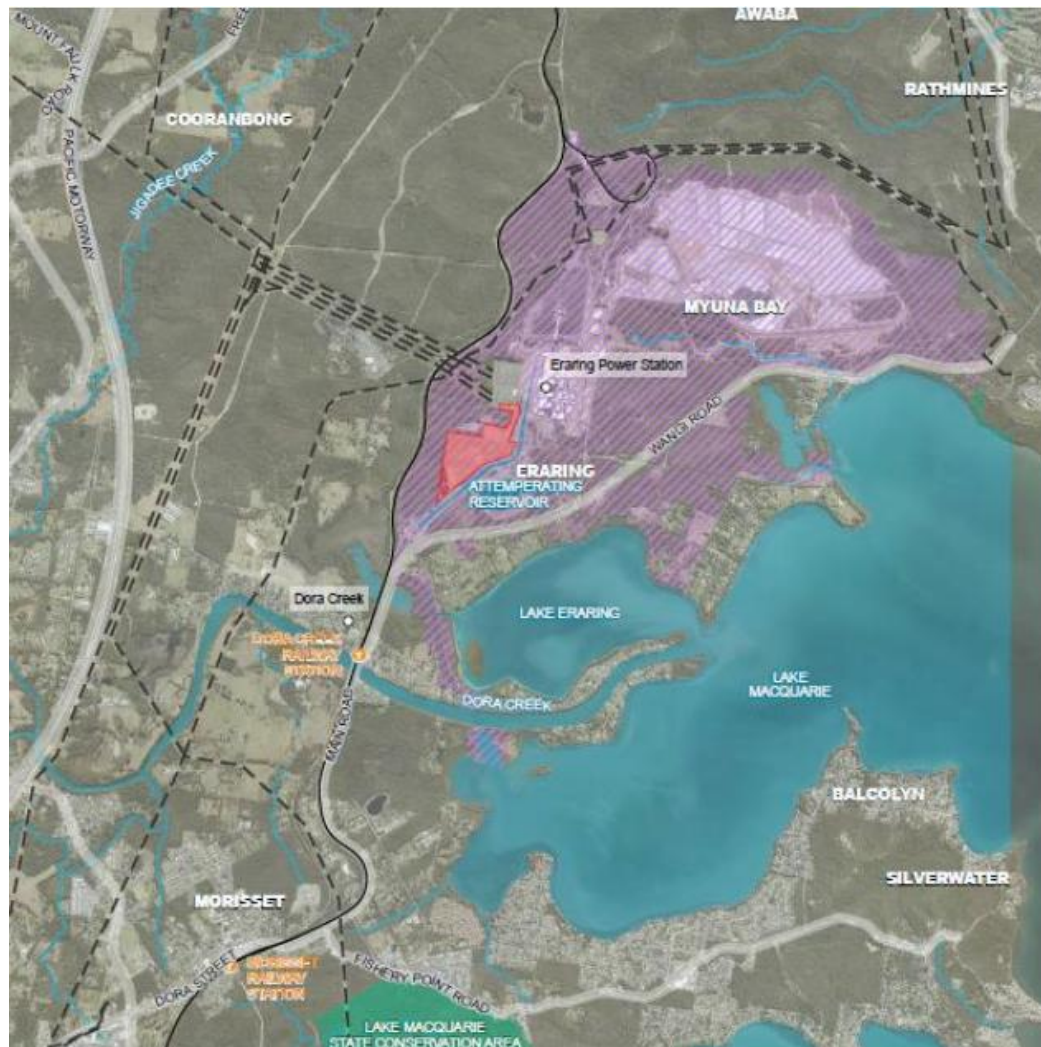


Figure 1: Eraring Power station location Map (Source: [NSW Department of Planning and Environment](#))

## Methodology

A combination of data sources and specialized software was used in this study to collect, process, and analyse emission and dispersion data for assessing pollution at Eraring Power Station. The use of software and tools, the origin of data, and the data input and analysis process are highlighted in this section.

### Software and tools:

**OpenLCA software:** Life-cycle assessment software, or LCA, is a specialised software that enables the analysis of comprehensive environmental impact. Emission data and modern environmental impacts can be processed through it, depending on the production of energy.

**SCREEN software:** The speed of pollutants is stimulated using this dispersion modelling tool. At different distances from the emission source, pollution concentration is estimated using this

software, depending on operational parameters including stack height and exit velocity, and exit temperature.

### **Data Sources:**

For collecting emission data, two data sources were used:

National Pollutant Inventory (NPI): A Variety of pollutant data, excluding carbon dioxide (CO<sub>2</sub>), is sourced from NPI.

Clean Energy Regulator (CER): The data gaps from NPI were covered by CER by obtaining specific CO<sub>2</sub> emissions from the electricity sectors for the years 2022-2023.

To achieve consistency, all data were gathered in a standardized unit of measurement.

Pollutant emission – kilograms (kgs)

CO<sub>2</sub> data – tons (t)

Electricity Production – megawatt-hours (MWh) for aligning the requirements of the OpenLCA model.

### **Data processing and analysis**

The raw data was formatted and standardized in Microsoft Excel according to the requirements of the OpenLCA software before being transferred to the OpenLCA software. Pollutant data were organized in kilograms (kg), CO<sub>2</sub> in tons (t), and output of electricity in megawatt-hours (MWh).

### **Data Import and Analysis on the OpenLCA software:**

- **Data Import:** The data processed in Excel was imported into the software.
- **Input/output parameters setup:** Pollutant emissions were assigned as input parameters, and the output of energy was assigned as output parameters. Over a defined period, the software represents the emissions and energy generation.
- **Creation of a product system and setup of calculations:** A functional unit of 1 MWh was established by creating a product system for analysis. For impact assessment, the following settings were specified:
- **Target Amount:** 1 MWh was set.
- **Methods of impact assessment:** For analyzing environmental impact across different categories, the ReCiPe 2016 Midpoint and Endpoint Hierarchies were selected.

### **SCREEN software for pollutant dispersion analysis**

Pollutant dispersion from the point source was stimulated using the SCREEN software, after emission impact analysis. This software includes the following data required:

- **Stack Height, Stack diameter, Exit Velocity, and Exit Temperature:** The emission source, physical and operational characteristics were identified using those parameters.

- **Measurement Distance:** The pollutant speed from the source to the target areas was determined by the distance at which dispersion rates were calculated.

Pollutant dispersion rates were modelled by entering those parameters into the software to estimate concentrations at different distances. A robust analysis of pollution dispersion and emission profiles is facilitated by this sequential approach, which is identical for both simple and complex terrain.

## Results and Data Presentation

The emissions and environmental performance of Eraring Power Station, situated on the western shore of Lake Macquarie, New South Wales, were analysed using 2023–24 data from the National Pollutant Inventory (NPI) and life-cycle impact modelling through OpenLCA applying the ReCiPe 2016 Midpoint (H) method. This section quantifies pollutant emissions, evaluates their likely impacts, and compares Eraring’s performance to the Australian average power generation mix (Tang and Mudd 2015; Asif et al. 2022).

### Pollutant Emission Rates

The NPI dataset shows Eraring released substantial amounts of combustion-related gases, particulates, and trace elements (Table 1). The most significant emissions were sulphur dioxide (SO<sub>2</sub>, 37 million kg) and oxides of nitrogen (NO<sub>x</sub>, 21.1 million kg), both critical for smog formation and acid rain (EPA NSW 2023). Carbon monoxide (CO, 1.65 million kg), hydrochloric acid (HCl, 810,000 kg), and particulate matter (PM<sub>10</sub> = 410,000 kg; PM<sub>2.5</sub> = 117,000 kg) represent further major outputs.

Table 1. NPI-reported annual emissions, Eraring Power Station (2023/24)

| Pollutant         | Total (kg) | Primary Environmental/Health Impact     |
|-------------------|------------|-----------------------------------------|
| SO <sub>2</sub>   | 37,000,000 | Acid deposition, respiratory irritation |
| NO <sub>x</sub>   | 21,100,000 | Photochemical smog, ozone precursor     |
| CO                | 1,651,000  | Toxic gas, incomplete combustion        |
| HCl               | 810,000    | Acid aerosols                           |
| PM <sub>10</sub>  | 410,000    | Respiratory irritation                  |
| PM <sub>2.5</sub> | 117,000    | Cardiovascular disease risk             |
| VOCs              | 207,200    | Ozone formation                         |
| Hg                | 28         | Neurotoxicity via bioaccumulation       |

|                               |        |                       |
|-------------------------------|--------|-----------------------|
| As                            | 31.5   | Carcinogenic effects  |
| Pb                            | 48.6   | Neurological toxicity |
| Other metals (Cd, Cr, Ni, Zn) | <1,000 | Ecotoxicity           |

Table 2. Relative emission ranking (normalised to highest pollutant = 1)

| Pollutant         | Relative Value | Rank |
|-------------------|----------------|------|
| SO <sub>2</sub>   | 1.00           | 1    |
| NO <sub>x</sub>   | 0.57           | 2    |
| CO                | 0.045          | 3    |
| HCl               | 0.022          | 4    |
| PM <sub>10</sub>  | 0.011          | 5    |
| PM <sub>2.5</sub> | 0.003          | 6    |

### Impact Modelling via OpenLCA

Table 3. ReCiPe 2016 Midpoint (H) impact results, Eraring Power Station

| Impact Category                   | Unit                    | Result | Principal Drivers                 |
|-----------------------------------|-------------------------|--------|-----------------------------------|
| Global warming                    | kg CO <sub>2</sub> eq   | 867.6  | CO <sub>2</sub> , NO <sub>x</sub> |
| Fine particulate formation        | kg PM <sub>2.5</sub> eq | 0.84   | PM, SO <sub>2</sub>               |
| Terrestrial acidification         | kg SO <sub>2</sub> eq   | 2.86   | SO <sub>2</sub> , NO <sub>x</sub> |
| Ozone formation (human health)    | kg NO <sub>x</sub> eq   | 1.34   | NO <sub>x</sub> , VOCs            |
| Human toxicity (carcinogenic)     | kg 1,4-DCB eq           | 0.029  | Cr VI, Ni, As                     |
| Human toxicity (non-carcinogenic) | kg 1,4-DCB eq           | 0.289  | Pb, Hg                            |
| Terrestrial ecotoxicity           | kg 1,4-DCB eq           | 15.48  | Heavy metals                      |

### Comparison with the Australian Average Generation

Table 4. Comparative emission intensities – Eraring vs Australian average

| Pollutant         | Eraring (t/MWh) | National Average (t/MWh) | Difference (%) |
|-------------------|-----------------|--------------------------|----------------|
| CO <sub>2</sub>   | 0.90            | 0.70                     | +29%           |
| SO <sub>2</sub>   | 0.015           | 0.011                    | +36%           |
| NO <sub>x</sub>   | 0.009           | 0.007                    | +28%           |
| PM <sub>2.5</sub> | 0.00005         | 0.00003                  | +40%           |



**Dispersion and Exposure Profiles**

Wind data show north-easterly flows transport pollutant plumes toward the south and south-west, influencing towns including Wyee and Morisset (EPA NSW 2023). Stable atmospheric conditions and nocturnal inversions lead to local fumigation events (Hannun and Razzaq 2022). Acidic gases (SO<sub>2</sub>, HCl) deposit into Lake Macquarie, altering pH and nutrient balance (Hunter Community Environment Centre 2023).

**Health and Climate Indicators**

Table 5. Estimated health and climate burden indicators

| Indicator                                      | Unit                                  | Estimated Value | Reference                   |
|------------------------------------------------|---------------------------------------|-----------------|-----------------------------|
| PM <sub>2.5</sub> -related hospital admissions | % increase per 10 µg m <sup>-3</sup>  | 6–10 %          | WHO 2021                    |
| Annual CO <sub>2</sub> emissions               | t CO <sub>2</sub> eq yr <sup>-1</sup> | ≈ 3,000,000     | Clean Energy Regulator 2023 |
| SO <sub>2</sub> acid deposition potential      | kg SO <sub>2</sub> eq                 | 2.86            | OpenLCA 2024                |
| Ecotoxicity potential                          | kg 1,4-DCB eq                         | 15.48           | OpenLCA 2024                |

**Discussion and Interpretation**

**Major Pollutants of Concern:**

Key pollutants include SO<sub>2</sub>, NO<sub>x</sub>, PM<sub>2.5</sub>, CO<sub>2</sub>, and Hg. SO<sub>2</sub> and NO<sub>x</sub> drive acidification and photochemical smog; PM<sub>2.5</sub> is linked to mortality; CO<sub>2</sub> contributes to climate change; Hg bioaccumulates in aquatic ecosystems (Asif et al. 2022; Vig et al. 2023).

**Dispersion and Receptor Exposure:**

Local meteorology and terrain enhance plume recirculation. Under calm conditions, ground-level concentrations rise, affecting nearby residential zones (EPA NSW 2022). Acidic deposition and trace-metal fallout influence soil fertility and aquatic biodiversity (Environmental Resource Management 2014).

**Comparison with the National Energy Mix:**

Eraring’s pollutant intensity remains above the national mean. Coal-based generation contributes over 85 % of SO<sub>2</sub> and NO<sub>x</sub> emissions, whereas gas and renewables produce < 10 % (Zhang 2020; Eguchi et al. 2021).

## Health and Ecosystem Impacts:

Table 6. Summary of key impact pathways

| Impact Type        | Primary Pollutants                                    | Key Effects                          | References                  |
|--------------------|-------------------------------------------------------|--------------------------------------|-----------------------------|
| Human health       | PM <sub>2.5</sub> , SO <sub>2</sub> , NO <sub>x</sub> | Respiratory & cardiovascular disease | WHO 2021; Webb et al. 2020  |
| Ecosystem          | Hg, As, Pb                                            | Bioaccumulation, toxicity            | Vig et al. 2023             |
| Climate            | CO <sub>2</sub>                                       | Global warming                       | IEA 2022                    |
| Soil acidification | SO <sub>2</sub> , NO <sub>x</sub>                     | Reduced fertility                    | Oyekale and Emagbetere 2022 |

## Uncertainties and Limitations:

NPI data reflect annual averages and omit temporal variability. The ReCiPe model assumes uniform dispersion and generic impact coefficients (Milousi et al. 2022). Health outcomes derive from population-based exposure–response functions, not direct monitoring. Nevertheless, the relative importance of pollutants is robust.

## Regulatory Compliance and Policy Context:

Eraring operates under the Protection of the Environment Operations (POEO) Act 2011 and Clean Air Regulation 2010, Group 5, with limits of 1,700 mg m<sup>-3</sup> for SO<sub>2</sub> and 1,500 mg m<sup>-3</sup> for NO<sub>x</sub> (EPA NSW 2022). Compliance is achieved, but total emissions remain high due to its 2,880 MW capacity. Ambient standards under the National Environment Protection Measure (NEPM 2021) set 25 µg m<sup>-3</sup> for PM<sub>2.5</sub> and 0.1 ppm for SO<sub>2</sub>; occasional exceedances occur during stable meteorology (EPA NSW 2023). Origin Energy’s planned retirement by 2032 will reduce these loads (NS Energy 2023).

## Recommendations

### 1. Flue Gas Desulphurisation (FGD):

The process of FGD is applied in eliminating the sulfur dioxide (SO<sub>2</sub>) emissions in the flue gases during combustion process of the fossil fuels, which is mainly coal. This is usually carried out through a wet scrubbing system whereby flue gas would be combined with lime (CaO) or limestone (CaCO<sub>3</sub>) and chemically reacted with sulfur dioxide to form calcium sulfate (CaSO<sub>3</sub>). It is safe to dispose of this by-product or even re-use it in construction as a drywall gypsum. (Eguchi, 2021)

### **Selective Catalytic Reduction (SCR):**

To minimize the emission of Nitrogen oxide ( $NO_x$ ) SCR technology is employed. During this process, reducing agent typically ammonia or urea gets injected on the flue gases, and the gases are passed through a catalytic converter. The catalyst helps in converting  $NO_x$  to the nitrogen ( $N_2$ ) and water ( $H_2O$ ) (Eguchi, 2021). SCR systems are highly efficient in suppressing  $NO_x$  up to 90%, leaving acid rain and ground-level ozone at bay.

## **2. Continuous Emission Monitoring System (CEMS)**

CEMS is the real-time monitoring system of air pollutants of  $PM_{2.5}$ ,  $SO_2$ ,  $NO_2$ , and  $Hg$  (Mercury). The system gives a continuous information on the concentration of these pollutants in the emissions from power plant or a factory. CEMS is usually composed of the following elements:

- a. **Sensors and analyzers:** Sensors and analyzers used to detect and measure the presence of certain pollutants such as particulate matter (PM), sulfur dioxide ( $SO_2$ ), nitrogen oxides ( $NO_x$ ), and mercury ( $Hg$ ).
- b. **Data collection and processing:** CEMS obtains data from sensors and transmit data to a central controllers unit to process it. This data is continuously being logged and analyzed in real-time. (EPA NSW, 2023)
- c. **Regulatory Compliance Reporting:** CEMS assists facilities to comply with the regulatory standards by constantly monitoring the level of emissions and ensure that they are not above the legally required levels. (EPA NSW, 2023)

## **3. Conformity of Regulatory Limits to the EU Industrial Emissions Directive (2010/75/EU)**

The EU Industrial Emission Directive (IED) establishes strict limits on the emission of pollutants by industrial plants, including the power plants. It controls the levels of emissions of  $SO_2$ ,  $NO_x$ ,  $PM$  (particulate matter), and mercury ( $Hg$ ). The directive aims to minimize pollution and encourage best practices in reducing the environmental impact of the industrial activity. Key features of IED compliance are:

- I. **Emission Limit Values (ELVs) :** The directive sets the maximum permissible level of various pollutants that can be emitted.
- II. **Best Available Techniques (BAT):** It obliges the facilities to use the most efficient method to reduce emissions with references to the technical feasibility, environmental impact, and cost (Romanello, 2022)

### **How It Works:**

- **Emissions monitoring:** Continuous emissions data, such as through CEMS, allows facilities to demonstrate compliance with the ELVs defined in the IED.

- **Implementation of BAT:** By adopting technologies like FGD and SCR, facilities not only are able to comply with the limit of emissions, but they also would be practicing the best techniques in reducing the pollution.

#### 4. **Battery Energy Storage System (BESS) and Decarbonisation of the NSW Grid.**

This is a very critical & important aspect of the shift towards renewable energy in New South Wales (NSW). BESS involves in utilization of mass lithium-ion batteries to store surplus power produced by renewable energies like wind and solar energy. This stored energy can be released to the grid when the demand is high or the renewable production is low, ensuring a constant and consistent power supply.

How It will Works:

- a) **Energy Storage:** When the renewable generation is excessive (e.g. on sunny or windy days), surplus energy is stored in batteries. This serves to prevent waste of energy.
- b) **Discharge:** When the demand is high or the renewable generation is low, the stored energy is released to the grid and therefore the supply is not lost when the fossil fuels are used.
- c) **Decarbonisation:** By facilitating the use of renewable energy, BESS reduces the need for fossil fuel-based power generation, contributing to the early decarbonisation of the grid.

## Conclusions

The current study evaluated the emissions and the environmental performance of the Eraring Power Station based on the certified data available on the Clean Energy Regulator (2023-24) and the National Pollutant Inventory (2023). In 2023/24, the facility had 13.55 million tonnes of CO<sub>2</sub>-equivalent emissions at an emission intensity of 0.868 t CO<sub>2</sub>-eq MWh<sup>-1</sup>, or about 40 percent above the average electricity generation intensity in Australia (CSIRO, 2022). It was found out that SO<sub>2</sub>, NO<sub>x</sub>, PM<sub>2.5</sub>, CO, and Hg are the major pollutants leading to acidity deposition, photochemical smog, and non-beneficial respiratory and cardiovascular health effects (WHO 2021; Farrow et al. 2020). Global warming potential, the formation of fine particulate matter, and terrestrial acidification were the most prevalent types of impact in the life-cycle modelling according to ReCiPe 2016 Midpoint (H) framework. Patterns of dispersion indicate that the ground-level pollutant concentrations sometimes surpass ambient limits of NEPM especially during the occurrence of a thermal inversion within the Lake Macquarie area (EPA NSW 2023). In as much as Eraring is operating within the stipulated emission limits, the volumes of its emissions are among the highest in the national level.

This report suggests the following:

1. Installation of Flue Gas Desulphurisation (FGD) and Selective Catalytic Reduction (SCR) equipment to control SO<sub>2</sub> and NO<sub>x</sub> emissions to 90 per cent (Eguchi et al. 2021).
2. Introduction of Continuous Emission Monitoring system (CEMS) on real time monitoring of PM<sub>2.5</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and Hg (EPA NSW, 2023).
3. Conformity of regulatory limits to the EU Industrial Emissions Directive (2010/75/EU) to the best practice internationally (Romanello et al. 2022).
4. Scheduling the shift to the Eraring Battery Energy Store System (BESS) to enable the early decarbonisation of the NSW grid (Clean Energy Regulator 2024).

To sum up, the activity of Eraring still has a quantifiable environmental and public-health impact. There is a need to upgrade emission-controlling technologies and accelerate the incorporation of renewable-energy to not only protect the air quality in the region, but also to facilitate long-term climate-reduction commitments of Australia.

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